Project 1: SRAM Cell Area Scaling Challenge

Project Summary

Problem Statement	• Baseline 6T SRAM bitcell must shrink 15-20% in area while remaining manufacturable. • Hard constraint: lithography resolution fixed at 22nm (cannot be relaxed). • Manufacturability must hold through Metal-2 (M2).
Selected Objective	• Reduce cell area by scaling 4 of 5 candidate parameters: Active area dimensions, Contact locations, Local Interconnect (LI) routing, Poly pitch. • Fin count intentionally left unchanged. • Study conducted on representative BB2 half-cell window; verified via layout (.cat) + process (.vproc) cross-analysis.

Baseline Status

Metric	Value	Metric	Value	Metric
BB2 cell size	216 x 172 nm	Poly pitch	86 nm	Fin count: 18 mandrels → 36 fins (SADP)
BB2 cell area	37,152 nm^2	Gate length	16 nm	Litho resolution: 22nm (fixed)

Final Recommendation — Scaling Attempts Summary

S.No	Parameter	Method	BB2 Result
1	Active area dimensions	Size-only, -10%, center-aligned	0% (enabler only)
2	Contact locations	Position-tighten, -10%	0%, but 13.3% contacts miss LI
3	LI routing	Size-only, -5% (10% breaches 22nm floor)	0% (enabler only)
4	Poly pitch	Y-axis scale, -10%	10.0% (first boundary-setting param)
5	All 4 combined	-10% uniform, X+Y, boundary redrawn	19.58% — FINAL RESULT

Recommendation: Adopt the combined 4-parameter scaling (Active dimensions + Contact locations + LI routing +Poly pitch, all -10%) as the final layout. This achieves 19.58% BB2 area reduction, within the 15-20% target, with Fin count preserved at baseline.

Residual risk: Gate length (14-15nm) sits below the fixed 22nm litho resolution floor — already marginal at baseline (16nm), flagged for litho-emulation verification during rebuild.